

NORTH SYDNEY GIRLS HIGH SCHOOL



2017 TRIAL HSC EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet has been provided
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I

Pages 3 - 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

Pages 7 – 13

60 Marks

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

NAME: _____ TEACHER: _____

STUDENT NUMBER: _____

QUESTION	MARK
1–10	/10
11	/15
12	/15
13	/15
14	/15
TOTAL	/70

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Consider the polynomial $P(x) = 3x^3 + 3x + a$.
If $x - 2$ is a factor of $P(x)$, what is the value of a ?
- (A) -30
- (B) -18
- (C) 18
- (D) 30
-
- 2 Let α , β and γ be the roots of $P(x) = 2x^3 - 5x^2 + 4x - 9$.
Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.
- (A) $\frac{5}{9}$
- (B) $-\frac{5}{9}$
- (C) $\frac{4}{9}$
- (D) $-\frac{4}{9}$

3 Which expression is equal to $\int \sin^2 2x \, dx$?

(A) $\frac{1}{2} \left(x - \frac{1}{4} \sin 4x \right) + c$

(B) $\frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right) + c$

(C) $\frac{1}{2} \left(x - \frac{1}{2} \sin 4x \right) + c$

(D) $\frac{1}{2} \left(x + \frac{1}{2} \sin 4x \right) + c$

4 Which of the following is equivalent to $\frac{\sin x}{1 - \cos x}$?

(A) $\tan \left(\frac{x}{2} \right)$

(B) $-\tan \left(\frac{x}{2} \right)$

(C) $\cot \left(\frac{x}{2} \right)$

(D) $-\cot \left(\frac{x}{2} \right)$

5 What are the asymptotes of $y = \frac{3x}{(x+1)(x-2)}$?

(A) $y = 0, x = -1, x = 2$

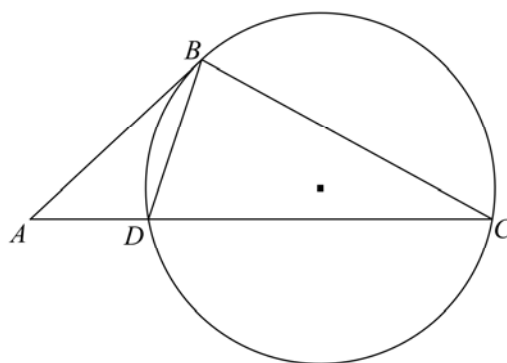
(B) $y = 0, x = 1, x = -2$

(C) $y = 3, x = -1, x = 2$

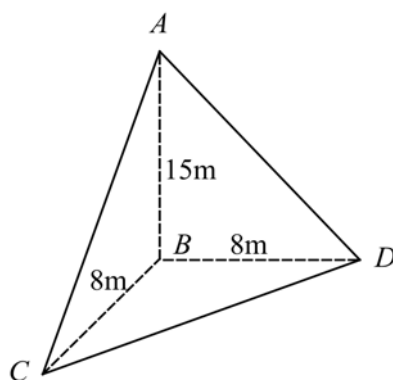
(D) $y = 3, x = 1, x = -2$

- 6 Which of the following is the range of the function $y = 2 \sin^{-1} x + \frac{\pi}{2}$?
- (A) $-\pi \leq y \leq \pi$
- (B) $-\pi \leq y \leq \frac{3\pi}{2}$
- (C) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
- (D) $-\frac{\pi}{2} \leq y \leq \frac{3\pi}{2}$
- 7 If P divides the interval AB internally in the ratio $m : n$, in what ratio does A divide the interval BP ?
- (A) $(m + n) : -n$
- (B) $(m + n) : -m$
- (C) $-n : (m + n)$
- (D) $-m : (m + n)$
- 8 What is a general solution of $\tan 2\theta \tan \theta = 1$?
- (A) $2n\pi \pm \frac{\pi}{3}$ where n is an integer.
- (B) $(6n \pm 1)\frac{\pi}{6}$ where n is an integer.
- (C) $(4n \pm 1)\frac{\pi}{6}$ where n is an integer.
- (D) $2n\pi \pm \frac{\pi}{6}$ where n is an integer.

- 9 In the diagram below, AB is the tangent to the circle at B and ADC is a straight line. If $AB : AD = 2 : 1$, then what is the ratio of the area of $\triangle ABD$ to the area of $\triangle CBD$?



- (A) 1:2
- (B) 1:3
- (C) 1:4
- (D) 2:3
- 10 In the figure below, AB is a vertical pole standing on horizontal ground BCD , where $\angle CBD = 90^\circ$. If the angle between the plane ACD and the horizontal ground is θ , then what is the value of θ closest to?



- (A) 45°
- (B) 53°
- (C) 62°
- (D) 69°

End of Section I

Section II

Total marks – 60

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 to 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Differentiate $\tan^{-1} \sqrt{x}$ with respect to x . **2**
- (b) Consider the function $f(x) = 1 + \frac{2}{x-3}$ for $x > 3$.
- (i) What is the range of $f(x)$? **1**
- (ii) Find the inverse function $f^{-1}(x)$ and state its domain. **1**
- (c) Use the substitution $u = 3 + x$ to find $\int \frac{x+1}{\sqrt{3+x}} dx$. **3**
- (d) Solve $\frac{4}{x+2} \geq \frac{1}{x}$. **3**
- (e) Find $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$. Show all working. **2**
- (f) (i) Neatly sketch the graph of $y = \sin^{-1} x$. **1**
- (ii) By considering areas on the graph in (i), find the exact value of $\int_0^{\frac{1}{2}} \sin^{-1} x \, dx$. **2**

End of Question 11

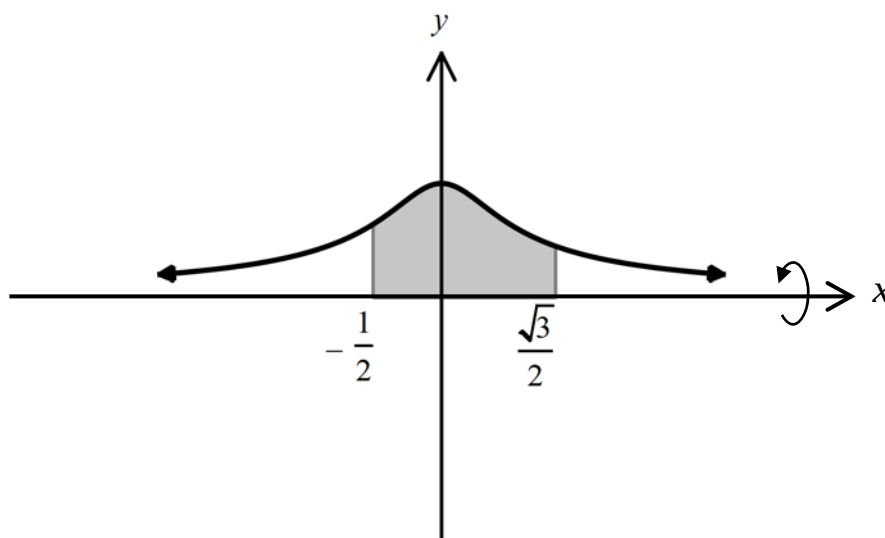
Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) During the early summer months, the rate of increase of the population P of cicadas is proportional to the excess of the population over 3000. The rate can be expressed by the differential equation $\frac{dP}{dt} = k(P - 3000)$ where t is the time in months and k is a constant. At the beginning of summer the population is 4000 and one month later it is 10 000.
- (i) Show that $P = 3000 + Ae^{kt}$ is a solution of the differential equation, 1
where A is a constant.
- (ii) Find the value of A . 1
- (iii) Show that the value of k is $\log_e 7$. 1
- (iv) After how many weeks will the population reach half a million? 2
(Assume 52 weeks in a year).
- (b) The angle between the line $4x + 3y = 8$ and the line $ax + by + c = 0$ is 45° . 3
Find the possible values of the ratio $a : b$.

Question 12 continues on page 9

- (c) The graph of $y = \frac{1}{\sqrt{1+4x^2}}$ is shown below.

3



The shaded region in the diagram is bounded by the curve $y = \frac{1}{\sqrt{1+4x^2}}$, the x -axis and the lines $x = -\frac{1}{2}$ and $x = \frac{\sqrt{3}}{2}$. Find the exact volume of the solid of revolution formed when the shaded region is rotated about the x -axis.

- (d) (i) Express $3\sin x + \sqrt{3}\cos x$ in the form $A\sin(x + \alpha)$, where $0 < \alpha < \frac{\pi}{2}$. 2
- (ii) Hence, or otherwise, sketch the graph of $y = 3\sin x + \sqrt{3}\cos x$ where $0 \leq x \leq 2\pi$. 2

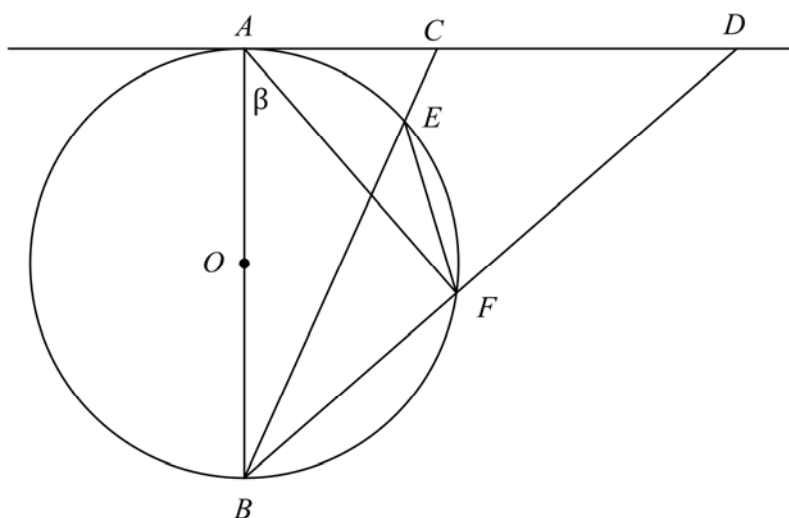
End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The acceleration of a particle as it moves in a straight line is given by $\frac{d^2x}{dt^2} = -12 \cos 2t$ where x is the displacement in metres of the particle from the origin at time t seconds. The particle starts from rest at the point $x = 3$.

- (i) Find the displacement, x , of the particle as a function of t . 2
- (ii) At what time is the particle at $x = 0$, and moving towards its initial position? 1

- (b) In the diagram below, the straight line ACD is a tangent at A to the circle with centre O . The interval AOB is a diameter of the circle. The intervals BC and BD meet the circle at E and F respectively. Let $\angle BAF = \beta$.



- (i) Explain why $\angle ABF = 90^\circ - \beta$. 1
- (ii) Prove that the quadrilateral $CDFE$ is cyclic. 3

Question 13 continues on page 11

- (c) A ball on a spring is moving in simple harmonic motion with a vertical velocity $v \text{ cms}^{-1}$ given by $v^2 = -8 + 24y - 4y^2$ where y is the vertical displacement in cm.
- (i) Find the acceleration of the ball in terms of y . 2
- (ii) Find the centre of motion of the ball. 1
- (iii) Find the period of the oscillation. 1
- (d) (i) Show that $n + (n + 1) + (n + 2) + \dots + (2n + 1) = \frac{(3n + 1)(n + 2)}{2}$ 1
- (ii) Hence prove by mathematical induction that for all integers $n \geq 1$, 3

$$1 + (2 + 3) + (3 + 4 + 5) + \dots + [n + (n + 1) + (n + 2) + \dots + (2n - 1)] = \frac{n^2}{2}(n + 1).$$

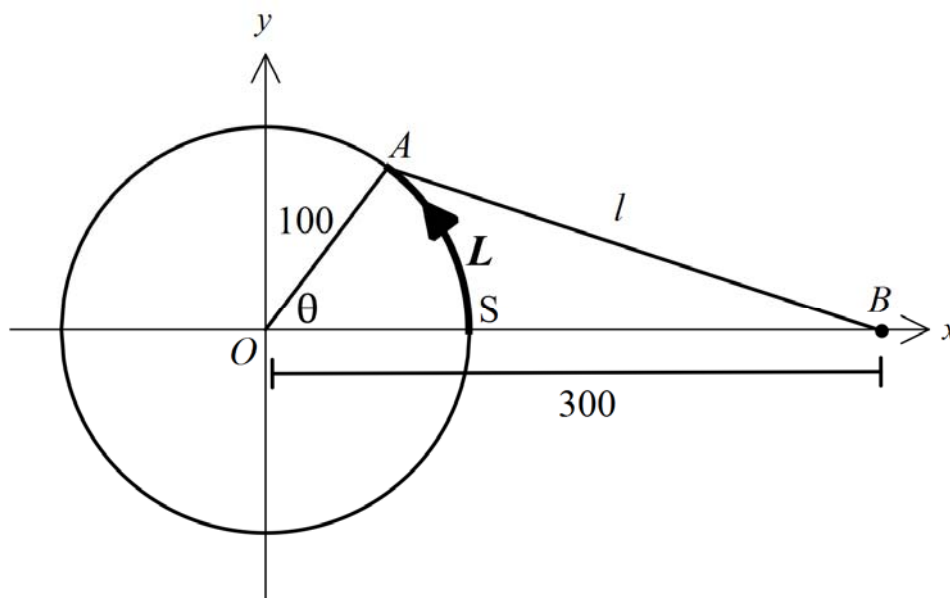
End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that the equation of the normal to the parabola $x^2 = 4y$ at the point $P(2p, p^2)$ is $x + py = 2p + p^3$. **2**
- (ii) S is the focus of the parabola $x^2 = 4y$ and T is a point on the normal such that ST is perpendicular to the normal. Write down the equation of ST . **1**
- (iii) Prove that the locus of T is a parabola and state its vertex and focal length. **3**
- (b) (i) Show that $1 + e^{-x} = \frac{e^x + 1}{e^x}$. **1**
- (ii) The velocity v of a particle moving along the x -axis is given by $\frac{dx}{dt} = 1 + e^{-x}$ where x is the displacement of the particle from the origin in metres. Initially the particle is at the origin. **3**
- Find the time taken by the particle to reach a velocity of $1\frac{1}{2} \text{ ms}^{-1}$.

Question 14 continues on page 13

- (c) A runner sprints in an anticlockwise direction around a circular track of radius 100 metres with centre O at a constant speed of 5 m/s. The runner's friend is standing at B , a distance of 300 metres from the centre of the track.



The runner starts at S and t seconds later is at point A . The distance AB between the two friends is l and the distance covered by the runner on the track is L . Let the angle subtended by the arc SA be θ .

- (i) From the diagram the coordinates of A are $(100 \cos \theta, 100 \sin \theta)$. 1
 Use the distance formula to show that $l = 100\sqrt{10 - 6 \cos \theta}$.
- (ii) At what rate is the distance between the friends changing at the moment 4
 when the runner is 250 metres from his friend and getting closer to him.

End of paper



Mathematics Extension 1

SOLUTIONS

- 1** Consider the polynomial $P(x) = 3x^3 + 3x + a$.
If $x - 2$ is a factor of $P(x)$, what is the value of a ?

Answer: A

$$P(2) = 0 \quad 3 \times 2^3 + 3(2) + a = 0$$

$$a = -30$$

- 2** Let α , β and γ be the roots of $P(x) = 2x^3 - 5x^2 + 4x - 9$.
Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

Answer: C

$$\frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{\frac{4}{2}}{\frac{9}{2}} = \frac{4}{9}$$

- 3** Which expression is equal to $\int \sin^2 2x \, dx$?

Answer: A

$$\int \sin^2 2x \, dx = \frac{1}{2} \int (1 - \cos 4x) \, dx$$

$$= \frac{1}{2} \left(x - \frac{1}{4} \sin 4x \right) + c$$

- 4** Which of the following is equivalent to $\frac{\sin x}{1 - \cos x}$?

Answer: C

$$\text{Let } t = \tan\left(\frac{x}{2}\right)$$

$$\frac{\frac{2t}{1+t^2}}{1 - \frac{1-t^2}{1+t^2}} = \frac{2t}{1+t^2 - 1+t^2}$$

$$= \frac{1}{t}$$

$$= \cot\left(\frac{x}{2}\right)$$

5 What are the asymptotes of $y = \frac{3x}{(x+1)(x-2)}$?

Answer: A

Vertical asymptotes at $x = -1, x = 2$

$$\lim_{x \rightarrow \infty} \frac{\frac{3x}{x^2}}{\frac{x}{x^2} - \frac{2}{x^2}} = \lim_{x \rightarrow \infty} \frac{\frac{3}{x}}{1 - \frac{2}{x}} \quad \therefore \text{horizontal asymptote at } y = 0$$

$$= 0$$

6 Which of the following is the range of the function $y = 2 \sin^{-1} x + \frac{\pi}{2}$?

Answer: D

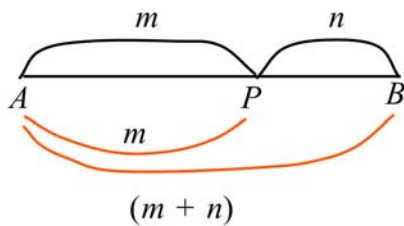
$$-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$$

$$-\pi \leq 2 \sin^{-1} x \leq \pi$$

$$-\frac{\pi}{2} \leq 2 \sin^{-1} x + \frac{\pi}{2} \leq \frac{3\pi}{2}$$

7 If P divides the interval AB internally in the ratio $m : n$, in what ratio does A divide the interval BP ?

Answer: B



8 What is a general solution of $\tan 2\theta \tan \theta = 1$?

Answer: B

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} \times \tan \theta = 1$$

$$2 \tan^2 \theta = 1 - \tan^2 \theta$$

$$3 \tan^2 \theta = 1$$

$$\tan \theta = \pm \frac{1}{\sqrt{3}} \Rightarrow \tan \theta = \tan \left(\pm \frac{\pi}{6} \right)$$

$$\theta = \pm \frac{\pi}{6} + n\pi \text{ where } n \text{ is an integer}$$

$$= (6n \pm 1) \frac{\pi}{6}$$

- 9** In the diagram below, AB is the tangent to the circle at B and ADC is a straight line. If $AB : AD = 2 : 1$, then what is the ratio of the area of $\triangle ABD$ to the area of $\triangle CBD$?

Answer: B

$$AB^2 = AD \times AC$$

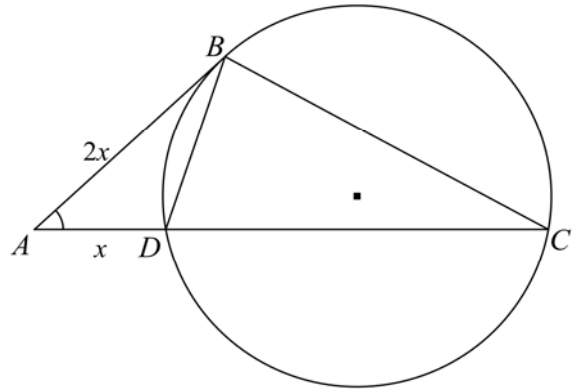
$$4x^2 = x \times AC \quad \Rightarrow AC = 4x$$

$\triangle ABD$ is similar to $\triangle ACB$ (two sides in ratio and included angle equal)

ratio of sides 1:2

ratio of areas 1:4

$\therefore$ area of $\triangle ABD$ to area of $\triangle CBD$ is 1:3



- 10** In the figure below, AB is a vertical pole standing on horizontal ground BCD , where $\angle CBD = 90^\circ$. If the angle between the plane ACD and the horizontal ground is θ , then what is the value of $\tan \theta$?

Answer: D

$$CD^2 = \sqrt{8^2 + 8^2} \quad \text{Pythagoras Theorem}$$

$$= 8\sqrt{2}$$

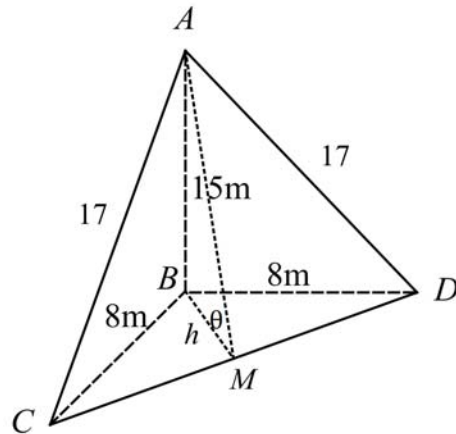
$$\text{Area of } \triangle BCD = \frac{1}{2} \times 8 \times 8$$

$$= 32$$

$$\text{Area of } \triangle BCD = \frac{1}{2} \times 8\sqrt{2} \times h$$

$$\therefore \frac{1}{2} \times 8\sqrt{2} \times h = 32 \quad \Rightarrow h = 4\sqrt{2}$$

$$\tan \theta = \frac{15}{4\sqrt{2}} \quad \Rightarrow \theta \approx 69^\circ$$



Question 11 (15 marks)

(a) Differentiate $\tan^{-1}\sqrt{x}$ with respect to x . 2

$$\begin{aligned}\frac{d}{dx}(\tan^{-1}\sqrt{x}) &= \frac{1}{1+x} \times \frac{1}{2\sqrt{x}} \\ &= \frac{1}{\sqrt{x}(2x+2)}\end{aligned}$$

(b) Consider the function $f(x) = 1 + \frac{2}{x-3}$ for $x > 3$.

(i) What is the range of $f(x)$? 1

(ii) Find the inverse function $f^{-1}(x)$ and state its domain. 1

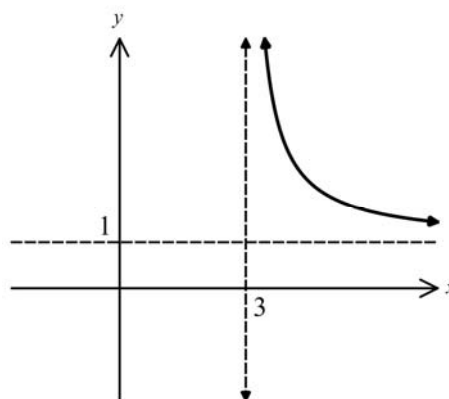
(i) $y > 1$

(ii) $x = 1 + \frac{2}{y-3}$

$$x-1 = \frac{2}{y-3}$$

$$y-3 = \frac{2}{x-1}$$

$$y = \frac{2}{x-1} + 3 \quad (\text{domain: } x > 1)$$

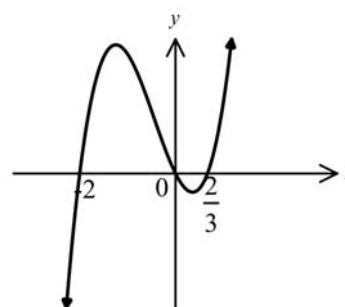


(c) Use the substitution $u = 3 + x$ to find $\int \frac{x+1}{\sqrt{3+x}} dx$. 3

$$\begin{aligned}\int \frac{x+1}{\sqrt{3+x}} dx &= \int \frac{u-2}{\sqrt{u}} du & u &= 3+x \\ &= \int \left(u^{\frac{1}{2}} - 2u^{-\frac{1}{2}} \right) du & du &= dx \\ &= \frac{2}{3} u^{\frac{3}{2}} - 2 \times 2u^{\frac{1}{2}} + C & x &= u-3 \\ &= \frac{2}{3} (3+x)^{\frac{3}{2}} - 4(3+x)^{\frac{1}{2}} + C & x+1 &= u-2\end{aligned}$$

(d) Solve $\frac{4}{x+2} \geq \frac{1}{x}$. 3

$$\begin{aligned}\frac{4}{x+2} &\geq \frac{1}{x} & x &\neq 0, 2 \\ 4(x+2)x^2 &\geq (x+2)^2 x \\ 4(x+2)x^2 - x(x+2)^2 &\geq 0 \\ x(x+2)[4x - (x+2)] &\geq 0 \\ x(x+2)(3x-2) &\geq 0 & x &\neq 0, 2 \\ \text{From the graph: } &-2 < x < 0, \quad x \geq \frac{2}{3}\end{aligned}$$



(e) Find $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$. Show all working.

2

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} &= \lim_{x \rightarrow 0} \frac{1 - (\cos^2 x - \sin^2 x)}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{(2 \sin^2 x)}{x^2} \\ &= 2 \times \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 \\ &= 2 \times \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \times \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \\ &= 2 \times 1 \times 1 \\ &= 2 \end{aligned}$$

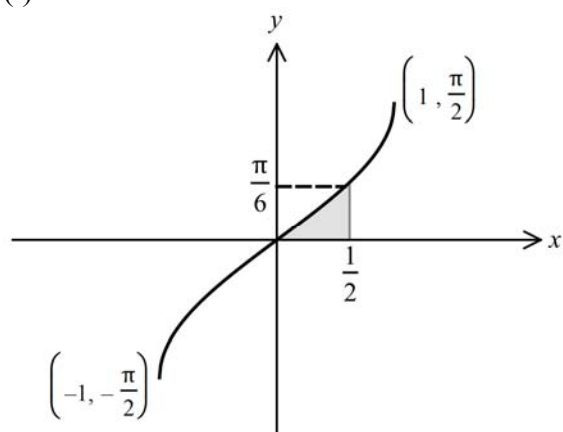
(f) (i) Neatly sketch the graph of $y = \sin^{-1} x$.

1

(ii) By considering the graph in (i), find the exact value of $\int_0^{\frac{1}{2}} \sin^{-1} x \, dx$.

2

(i)



$$\begin{aligned} \text{(ii)} \quad \int_0^{\frac{1}{2}} \sin^{-1} x \, dx &= \frac{\pi}{6} \times \frac{1}{2} - \int_0^{\frac{\pi}{6}} \sin y \, dy \\ &= \frac{\pi}{12} + \left[\cos y \right]_0^{\frac{\pi}{6}} \\ &= \frac{\pi}{12} + \left(\frac{\sqrt{3}}{2} - 1 \right) \\ &= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1 \end{aligned}$$

Question 12 (15 marks)

- (a) During the early summer months, the rate of increase of the population P of cicadas is proportional to the excess of the population over 3000. The rate can be expressed by the differential equation $\frac{dP}{dt} = k(P - 3000)$ where t is the time in months and k is a constant. At the beginning of summer the population is 4000 and one month later, it is 10 000.
- (i) Show that $P = 3000 + Ae^{kt}$ is a solution of the differential equation, 1
where A is a constant.
- (ii) Find the value of A . 1
- (iii) Show that the value of k is $\log_e 7$. 1
- (iv) After how many weeks will the population reach half a million? 2
(Assume 52 weeks in a year).

(a) (i) $P = 3000 + Ae^{kt} \Rightarrow P - 3000 = Ae^{kt}$

$$\begin{aligned} \text{LHS} &= \frac{dP}{dt} \\ &= kAe^{kt} \\ &= k(P - 3000) \\ &= \text{RHS} \end{aligned}$$

$\therefore P = 3000 + Ae^{kt}$ is a solution.

(ii) When $t = 0$, $P = 4000$

$$4000 = 3000 + A$$

$$A = 1000$$

(iii) When $P = 10\,000$ and $t = 1$, $10\,000 = 3000 + 1000e^k$

$$7000 = 1000e^k$$

$$e^k = 7$$

$$k = \log_e 7$$

(iv) $P = 500\,000$

$$500\,000 = 3000 + 1000e^{t \log_e 7}$$

$$497\,000 = 1000e^{t \log_e 7}$$

$$e^{t \log_e 7} = 497$$

$$t \log_e 7 = \log_e 497$$

$$t = \frac{\log_e 497}{\log_e 7} \approx 3.19$$

$$\frac{3.19}{12} \times 52 \approx 13.83 \text{ weeks}$$

It will take 14 weeks.

- (b) The angle between the line $4x + 3y = 8$ and the line $ax + by + c = 0$ is 45° . 3
Find the possible values of the ratio $a : b$.

$$m_1 = -\frac{4}{3} \quad m_2 = -\frac{a}{b}$$

$$\tan 45^\circ = \left| \frac{-\frac{4}{3} - \left(-\frac{a}{b}\right)}{1 + \frac{4}{3} \times \frac{a}{b}} \right|$$

$$1 = \left| \frac{-4b + 3a}{3b + 4a} \right|$$

$$1 = \frac{|-4b + 3a|}{|3b + 4a|}$$

$$\therefore -4b + 3a = 3b + 4a \quad \text{or} \quad -4b + 3a = -(3b + 4a)$$

$$a = -7b$$

$$7a = b$$

$$\frac{a}{b} = -7$$

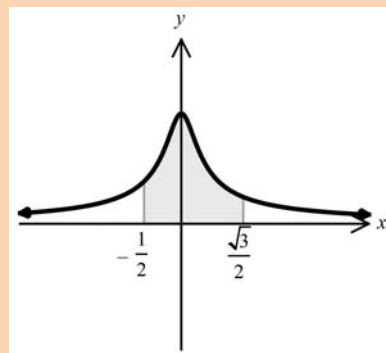
$$\frac{a}{b} = \frac{1}{7}$$

Ratio of $a : b$ is $-7:1$ or $1:7$

(c) The graph of $y = \frac{1}{\sqrt{1+4x^2}}$ is shown below.

3

The shaded region in the diagram is bounded by the curve $y = \frac{1}{\sqrt{1+4x^2}}$, the x -axis and the lines $x = -\frac{1}{2}$ and $x = \frac{\sqrt{3}}{2}$. Find the volume of the solid of revolution formed when the shaded region is rotated about the x -axis.



$$\begin{aligned}
 V &= \pi \int_a^b y^2 dx \\
 &= \pi \int_{-\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{1}{1+4x^2} dx \\
 &= \frac{\pi}{4} \int_{-\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{1}{\left(\frac{1}{2}\right)^2 + x^2} dx \\
 &= \frac{\pi}{4} \times 2 \left[\tan^{-1} 2x \right]_{-\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \\
 &= \frac{\pi}{2} \left[\tan^{-1} \sqrt{3} - \tan^{-1}(-1) \right] \\
 &= \frac{\pi}{2} \left[\frac{\pi}{3} - \left(-\frac{\pi}{4} \right) \right] \\
 &= \frac{7\pi^2}{24} u^3
 \end{aligned}$$

- (d) (i) Express $3\sin x + \sqrt{3}\cos x$ in the form $A\sin(x+\alpha)$, where $0 < \alpha < \frac{\pi}{2}$. **2**
- (ii) Hence, or otherwise, sketch the graph of $y = 3\sin x + \sqrt{3}\cos x$ where $0 \leq x \leq 2\pi$. **2**

(i)

$$3\sin x + \sqrt{3}\cos x = A\sin x \cos \alpha + A\cos x \sin \alpha$$

$$\text{equating coefficients: } A\cos \alpha = 3 \quad A\sin \alpha = \sqrt{3}$$

$$A^2 \cos^2 \alpha + A^2 \sin^2 \alpha = 12$$

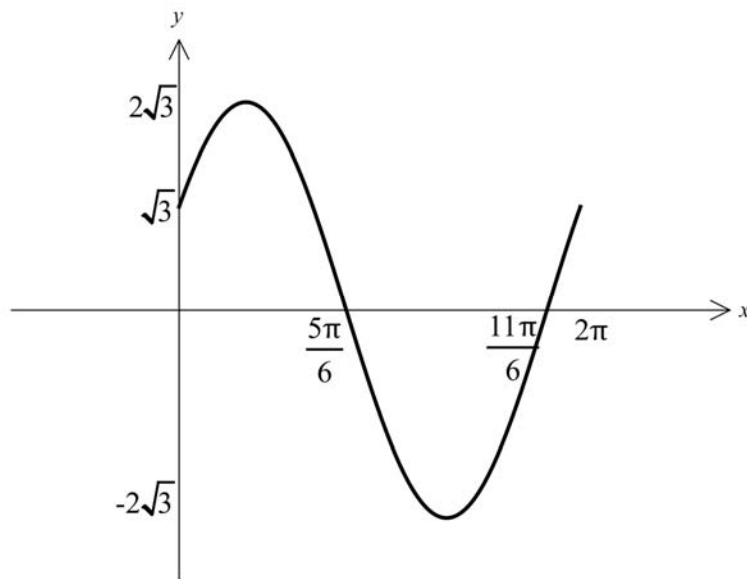
$$A^2 = 12 \quad \Rightarrow \quad A = 2\sqrt{3}$$

$$\text{Also, } \frac{A\sin \alpha}{A\cos \alpha} = \frac{\sqrt{3}}{3}$$

$$\tan \alpha = \frac{1}{\sqrt{3}} \quad \Rightarrow \quad \alpha = \frac{\pi}{6}$$

$$3\sin x + \sqrt{3}\cos x = 2\sqrt{3}\sin\left(x + \frac{\pi}{6}\right)$$

(ii)



Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The acceleration of a particle as it moves in a straight line is given by $\frac{d^2x}{dt^2} = -12 \cos 2t$ and the particle started from rest at the point $x = 3$.
- (i) Find the displacement, x , of the particle as a function of t . **2**
- (ii) At what time is the particle first at $x = 0$, and moving towards its initial position? **1**

(i) $\ddot{x} = -12 \cos 2t$

$$\dot{x} = -\frac{12 \sin 2t}{2} + c$$

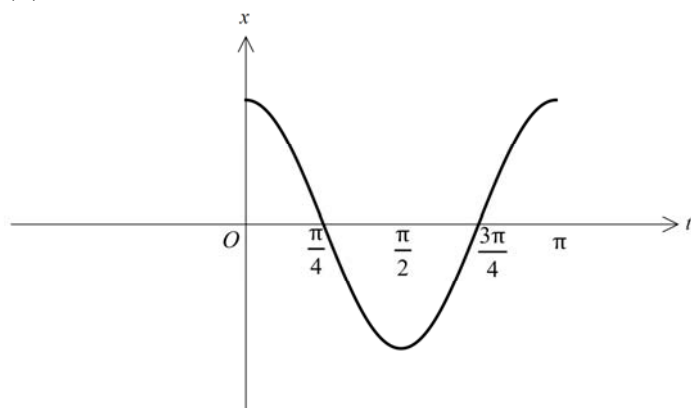
$$= -6 \sin 2t + c \quad \text{when } t = 0, v = 0 \Rightarrow c = 0$$

$$\dot{x} = -6 \sin 2t$$

$$x = \frac{6 \cos 2t}{2} + c_1 \quad \text{when } t = 0, x = 3 \Rightarrow 3 = 3 + c_1 \therefore c_1 = 0$$

$$x = 3 \cos 2t$$

(ii)



From the graph, $x = 0$ when particle is moving towards its initial position at $t = \frac{3\pi}{4}$ seconds.

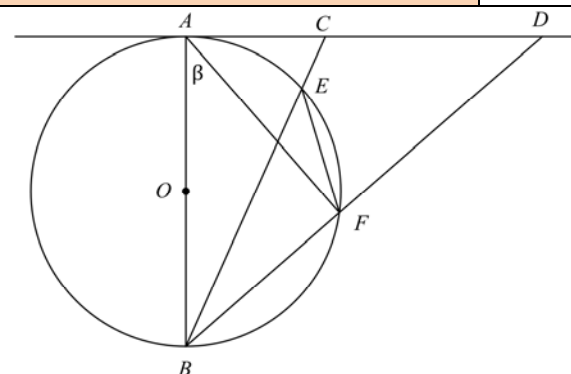
- (b) In the diagram below, the straight line ACD is a tangent at A to the circle with centre O . The interval AOB is a diameter of the circle. The intervals BC and BD meet the circle at E and F respectively. Let $\angle BAF = \beta$.

- (i) Explain why $\angle ABF = 90^\circ - \beta$. **1**
- (ii) Prove that the quadrilateral $CDFE$ is cyclic. **3**

- (i) $\angle AFB = 90^\circ$ (angle in a semi-circle)
 $\angle ABF + \beta + 90^\circ = 180^\circ$ (angle sum of triangle)
 $\angle ABF = 90 - \beta$

- (ii) $\angle BAF = \angle BEF$ (angle in the same segment)
 $= \beta$
 $\angle BAD = 90^\circ$ (tangent perpendicular to radius)
 In $\triangle BAD$, $\angle ADB + (90 - \beta) + 90 = 180$
 $\angle ADB = \beta$

$\angle CDF = \angle FEB = \beta$
 $\therefore CDFE$ is cyclic (exterior angle of a cyclic quadrilateral)



- (c) A ball on a spring is moving in simple harmonic motion with a vertical

velocity $v \text{ cms}^{-1}$ given by $v^2 = -8 + 24y - 4y^2$ where y is the vertical displacement in cm.

- | | | |
|-------|---|----------|
| (i) | Find the acceleration of the ball in terms of y . | 2 |
| (ii) | Find the centre of motion of the ball. | 1 |
| (iii) | Find the period of the oscillation. | 1 |

(i)

$$v^2 = -8 + 24y - 4y^2$$

$$\frac{1}{2}v^2 = -4 + 12y - 2y^2$$

$$\therefore \ddot{y} = \frac{d}{dy}(-4 + 12y - 2y^2)$$

$$= 12 - 4y$$

$$= -4(y - 3)$$

- (ii) centre of motion is at $x = 3$.

iii)

$$n^2 = 4$$

$$n = 2$$

$$\therefore T = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi \text{ sec}$$

- | | | | |
|-----|------|---|----------|
| (d) | (i) | Show that $n + (n+1) + (n+2) + \dots + (2n+1) = \frac{(3n+1)(n+2)}{2}$ | 1 |
| | (ii) | Hence prove by mathematical induction that for all integers $n \geq 1$,
$1 + (2+3) + (3+4+5) + \dots + [n + (n+1) + (n+2) + \dots + (2n-1)] = \frac{n^2}{2}(n+1)$. | 3 |

- (i) arithmetic series: $a = n$, $l = (2n+1)$, number of terms $= (2n+1) - n + 1$
 $= n+2$

$$\begin{aligned} \therefore n + (n+1) + (n+2) + \dots + (2n+1) &= \frac{(n+2)}{2}(n + (2n+1)) \\ &= \frac{(3n+1)(n+2)}{2} \end{aligned}$$

- (ii) Prove true for $n=1$

$$\text{LHS} = 1 \quad \text{RHS} = \frac{1}{2}(1+1) = 1$$

$$\text{LHS} = \text{RHS} \therefore \text{true for } n = 1.$$

Assume true for $n = k$.

$$1 + (2+3) + (3+4+5) + \dots + [k + (k+1) + (k+2) + \dots + (2k-1)] = \frac{k^2}{2}(k+1).$$

Prove true for $n = k+1$

Required to prove:

$$1 + (2+3) + \dots + [k + (k+1) + \dots + (2k-1)] + [(k+1) + ((k+1)+1) + \dots + (2(k+1)-1)] = \frac{(k+1)^2}{2}((k+1)+1).$$

$$1 + (2+3) + \dots + [k + (k+1) + \dots + (2k-1)] + [(k+1) + (k+2) + \dots + (2k+1)] = \frac{(k+1)^2}{2}(k+2)$$

$$\text{LHS} = 1 + (2+3) + \dots + [k + (k+1) + \dots + (2k-1)] + [(k+1) + (k+2) + \dots + (2k+1)]$$

$$\begin{aligned}
&= \frac{k^2}{2}(k+1) + [(k+1) + (k+2) + \dots + (2k+1)] \text{ from assumption} \\
&= \frac{k^2}{2}(k+1) + \frac{(3k+1)(k+2)}{2} - k \quad \text{from part (i)} \\
&= \frac{k^2(k+1) + 3k^2 + 7k + 2 - 2k}{2} \\
&= \frac{k^2(k+1) + (3k+2)(k+1)}{2} \\
&= \frac{(k+1)[k^2 + 3k + 2]}{2} \\
&= \frac{(k+1)(k+1)(k+2)}{2} \\
&= \frac{(k+1)^2(k+2)}{2} \\
&= \text{RHS}
\end{aligned}$$

Hence, if the result is true for $n = k$, then it is true for $n = k + 1$

$\therefore$ the result is true for all $n \geq 1$ by mathematical induction

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a)	(i)	Show that the equation of the normal to the parabola $x^2 = 4y$ at the point $P(2p, p^2)$ is $x + py = 2p + p^3$.	2
	(ii)	S is the focus of the parabola $x^2 = 4y$ and T is a point on the normal such that ST is perpendicular to the normal. Write down the equation of ST .	1
	(iii)	Prove that the locus of T is a parabola with vertex $(0,1)$ and with focal length $\frac{1}{4}$ that of the parabola $x^2 = 4y$.	3

(i)

$$x^2 = 4y \quad (\text{focal length } a = 1)$$

$$y = \frac{x^2}{4}$$

$$\frac{dy}{dx} = \frac{x}{2}$$

At P , $x = 2p$, $\frac{dy}{dx} = \frac{2p}{2} = p$ (gradient of tangent at P)

$$\therefore \text{Gradient of normal} = -\frac{1}{p}$$

Equation of normal:

$$y - y_1 = m(x - x_1)$$

$$y - p^2 = -\frac{1}{p}(x - 2p)$$

$$py - p^3 = -x + 2p$$

$$x + py = 2p + p^3$$

(ii)

Gradient of $ST = p$

Equation of ST : $y = px + 1$

(iii)

Since T is the intersection of ST and PT

$$\left. \begin{array}{l} y = px + 1 \end{array} \right\} \dots\dots\dots(1)$$

$$\left. \begin{array}{l} x + py = 2p + p^3 \end{array} \right\} \dots\dots\dots(2)$$

$$\underline{py = p^2x + p} \dots\dots\dots 1 \times p \dots\dots(3)$$

$$x = p + p^3 - p^2x \dots\dots(2) - (3)$$

$$x(1 + p^2) = p(1 + p^2)$$

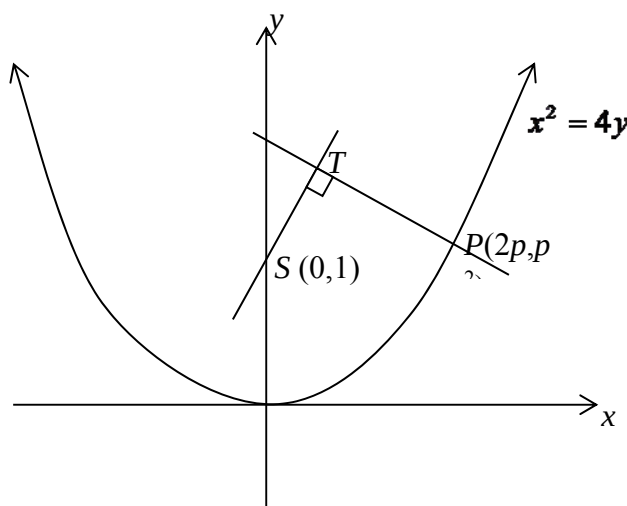
$$x = p$$

$$y = p^2 + 1$$

$\therefore$ Cartesian equation of T :

$$y = x^2 + 1, \text{ i.e. } x^2 = y - 1$$

Vertex $(0,1)$, focal length $a = \frac{1}{4}$



(b) (i) Show that $1 + e^{-x} = \frac{e^x + 1}{e^x}$. 1

(ii) The velocity v of a particle moving along the x -axis is given by 3
 $\frac{dx}{dt} = 1 + e^{-x}$ where x is the displacement of the particle from the origin in metres. Initially the particle is at the origin.

Find the time taken by the particle to reach a velocity of $1\frac{1}{2} \text{ ms}^{-1}$.

(i)
$$\text{RHS} = \frac{e^x + 1}{e^x}$$

$$= \frac{e^x}{e^x} + \frac{1}{e^x}$$

$$= 1 + e^{-x}$$

$$= \text{LHS}$$

(ii)
$$\frac{dx}{dt} = 1 + e^{-x}$$

$$\frac{dt}{dx} = \frac{1}{1 + e^{-x}}$$

$$= \frac{e^x}{1 + e^x}$$

$$t = \int \frac{e^x}{1 + e^x} dx = \log(1 + e^x) + c$$

$$\text{At } t = 0, x = 0 \quad 0 = \log_e(1 + 1) + c \quad \Rightarrow c = -\log_e 2$$

$$t = \log_e(1 + e^x) - \log_e 2$$

$$= \log_e \left(\frac{1 + e^x}{2} \right)$$

$$\text{when } v = \frac{3}{2}, \quad \frac{3}{2} = 1 + e^{-x}$$

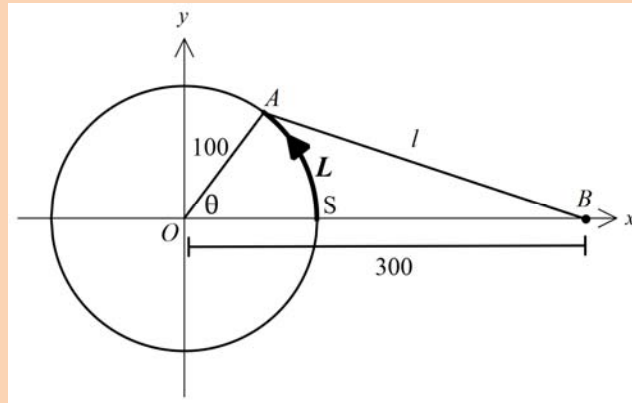
$$e^{-x} = \frac{1}{2} \Rightarrow e^x = 2 \Rightarrow x = \log_e 2$$

$$\text{when } x = \log_e 2, \quad t = \log_e(1 + e^{\log_e 2}) - \log_e 2$$

$$= \log_e \frac{3}{2}$$

It will take $\log_e \left(\frac{3}{2} \right)$ seconds.

- (c) A runner sprints in an anticlockwise direction around a circular track of radius 100 metres with centre O at a constant speed of 5 m/s. The runner's friend is standing at B , a distance of 300 metres from the centre of the track.



The runner starts at S and t seconds later is at point A . The distance AB between the two friends is l and the distance covered by the runner on the track is L . Let the angle subtended by the arc SA be θ .

- (i) From the diagram the coordinates of A are $(100\cos\theta, 100\sin\theta)$. 1

Use the distance formula to show that $l = 100\sqrt{10 - 6\cos\theta}$.

- (ii) At what rate is the distance between the friends changing at the moment when the runner is 250 metres from his friend and getting closer to him. 4

(i)

$$A (100\cos\theta, 100\sin\theta) \quad B (300, 0)$$

$$\begin{aligned} l^2 &= (100\cos\theta - 300)^2 + (100\sin\theta - 0)^2 \\ &= 100^2 \cos^2\theta - 2 \times 100 \times 300 \cos\theta + 300^2 + 100\sin^2\theta \\ &= 100^2 (\cos^2\theta + \sin^2\theta) + 300^2 - 2 \times 100 \times 300 \cos\theta \\ l &= \sqrt{100000 - 60000\cos\theta} \\ &= 100\sqrt{10 - 6\cos\theta} \end{aligned}$$

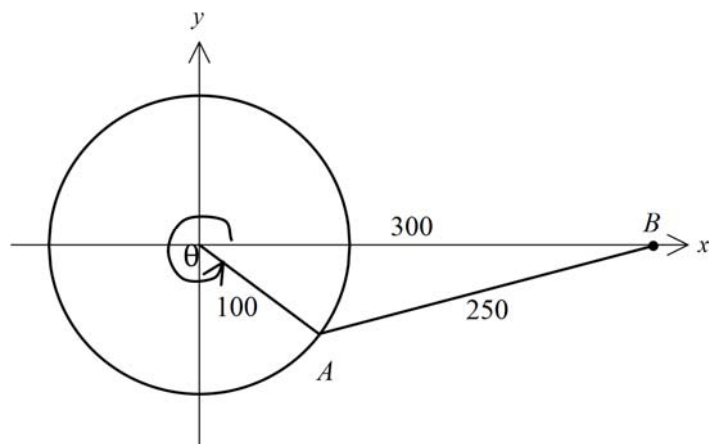
(ii) $\frac{dL}{dt} = 5 \quad \frac{dl}{d\theta} = \frac{600\sin\theta}{2\sqrt{10 - 6\cos\theta}}$

$$L = 100\theta \text{ (arc length)} \Rightarrow \frac{dL}{d\theta} = 100$$

$$\begin{aligned} \frac{dl}{dt} &= \frac{dl}{d\theta} \times \frac{d\theta}{dL} \times \frac{dL}{dt} \\ &= \frac{600\sin\theta}{2\sqrt{10 - 6\cos\theta}} \times \frac{1}{100} \times 5 \\ &= \frac{15\sin\theta}{\sqrt{10 - 6\cos\theta}} \end{aligned}$$

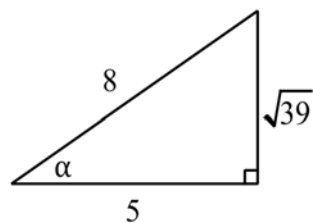
$$\text{When } l = 250, \quad 250 = 100\sqrt{10 - 6\cos\theta}$$

$$\left(\frac{5}{2}\right)^2 = 10 - 6\cos\theta \Rightarrow \cos\theta = \frac{5}{8}$$



θ is in the 4th quadrant.

$$\cos \theta = \frac{5}{8}, \quad \sin \theta = -\frac{\sqrt{39}}{8}$$



$$\begin{aligned} \frac{dl}{dt} &= \frac{15 \left(-\frac{\sqrt{39}}{8} \right)}{\sqrt{10 - 6 \times \frac{5}{8}}} \\ &= -\frac{\left(\frac{15\sqrt{39}}{8} \right)}{\sqrt{\frac{25}{4}}} = -\frac{15\sqrt{39}}{8} \times \frac{2}{5} \\ &= -\frac{3\sqrt{39}}{4} \end{aligned}$$

The distance between the two friends is decreasing a rate of $\frac{3\sqrt{39}}{4}$ ms ≈ 4.7 ms